

Jonas Kemnitzer

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EDUCATION

UAS HOF

M.Sc. in Computer Science
August 2023 | Hof, Germany

KEIMYUNG UNIVERSITY EXCHANGE SEMESTER

September 2019 | Daegu, S. Korea

UAS HOF

B.Sc. in Computer Science
March 2021 | Hof, Germany

COURSEWORK

GRADUATE

Modern Software Architecture
Information Systems Security
Data Mining & Machine Learning,
IT & Data Protection Law
Deep Learning Architectures

UNDERGRADUATE

Software Engineering
Algorithms & Data-Structures
Math & Statistics for CS
Applied Machine Learning
RESTful Web Services
Robotics & IT-Security

SKILLS

TECHNICAL SKILLS

- Python • PyTorch • OOP
- Git • Software Engineering
- SQL-DB's • Vector-DB's
- vLLM • Ollama • MLFlow
- Docker • Hugging Face Ecosystem
- $\text{\LaTeX}$ • TransformerLens
- Proxmox (VMs, LXC, Network)
- Self-hosted Infrastructure

SOFT SKILLS

- Agile Workflows
- Trilingual (German, English, Korean)
- Cross-Team Working
- Analytical Thinking & Adaptability

CLUBS

- Students Union • Intercultural Teacher • Freshman Mentor (Newcomer Days)
- Tennis Trainer (TC Hof)

WORK EXPERIENCE

IISYS AT HOF UNIVERSITY | RESEARCH ASSOCIATE & PHD-TRACK

RESEARCHER

August 2023 – Now | Hof, Germany

- Researched and developed multimodal AI systems within the Intelligent and Learning Systems group, covering the full stack from multi-GPU model training and fine-tuning to production inference deployment via vLLM and Ollama.
- Bridged academic research and industry application by transferring AI solutions into active production environments for small and medium-sized German businesses, translating published findings into practical, deployable systems.
- Conducted research in mechanistic interpretability, applying circuit analysis techniques to locate and characterise attention heads causally responsible for prompt injection in instruction-tuned LLMs.

IBM GERMANY | AI PROTOTYPE ENGINEER INTERN

April 2020 – August 2020 | Munich, Germany

- Developed anomaly detection pipelines on large-scale infrastructure datasets for road and bridge safety analysis using *Python* and *Apache Spark* on IBM Cloud.
- Designed and deployed a customer-facing conversational AI system on a *Pepper* robot, integrating *IBM Watson Assistant* with live APIs (weather, news) and custom dialogue flows for lobby guest interaction.
- Collaborated with solution architects to prototype and refine customer-oriented AI demos across multiple client engagements.

PROJECTS

EXPLAINABLE AI (LIME) | RESEARCH

Implemented and analysed the *Local Interpretable Model-agnostic Explanations (LIME)* algorithm in *Python*, evaluating input parameter sensitivity on the YouTube Comments Spam Dataset to reduce data complexity through systematic parameter optimisation.

SENTIMENT ANALYSIS USING NLP | RESEARCH

Built a proof-of-concept sentiment analysis pipeline for an online retail use case using *Python*, *TensorFlow*, and *Keras*, benchmarking multiple frameworks for performance trade-offs and refactoring the codebase to reduce overall runtime overhead.

PUBLICATIONS

- [1] J. Kemnitzer and C. Groth. Enhancing fitness visualization: application and efficacy of realistic inpainting techniques using diffusion models. In *International Conference on AI-Generated Content (AIGC 2024)*, volume 13649, page 1364907. International Society for Optics and Photonics, SPIE, 2025.
- [2] J. Kemnitzer and C. Groth. Beyond shared embedding spaces: Towards LLM architectures with explicit authority control for prompt security. In *2026 International Conference on Advances in Artificial Intelligence and Machine Learning (AAIML)*, Tokyo, Japan, Mar. 2026. IEEE. To appear.